

# Sonok Mahapatra

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## EDUCATION

### University of Maryland, College Park

Expected December 2027

*M.S & B.S Applied Math + B.S. Computer Science; Minor in Computational Finance*

GPA: 3.7

**Relevant Coursework:** Grad High Performance Computing, Grad Stochastic Processes, Grad Signal Processing, Probability Theory, Advanced Statistics, PDEs, Portfolio Management, Financial Markets, Machine Learning, Applied Linear Algebra, Computer Systems

## AWARDS & CERTIFICATIONS

Putnam Top 700 Scorer (14/120); Five-time AIME Qualifier ('18-'22); Jane Street Estimation Challenge Top 3; AMC Score (132/150); Bridgewater Prediction Market Competition (Top 35 World); Dean's List; 2nd Place Lockheed Martin CodeQuest; Presidential Scholar; Akuna Options 101 & Options Market Making 201; [Bloomberg Financial Fundamentals Certification](#); [AWS Certified Cloud Practitioner](#)

## EXPERIENCE

**Shopify – ML Platform** (Sept 2026 – Dec 2026) — Building ML infra for real-time auction and pricing systems serving 5M+ merchants

### Amazon Web Services

June 2026 – Present

*Software Development Engineer, AWS Security*

- Developed a sandbox environment for Rubicon, AWS Security's document processing platform, enabling isolated validation of infrastructure changes across security data pipelines handling 3K+ events/sec or 200M+ events/day
- Built production-fidelity testing infrastructure for Rubicon, replaying historical security workloads to validate distributed system behavior and accelerate outage recovery.
- Reducing pipeline validation and recovery time from 6+ hours to under 1 hour by eliminating shared-environment bottlenecks through automated pre-production testing.

### ProBanker

May 2025 – Aug 2025

*Software Engineering Intern*

- Built features for [Probanker's](#) multi-tenant SaaS platform simulating live financial markets for universities and corporate clients.
- Developed responsive, production-grade UIs in Angular with REST API integrations and modular frontend architecture.
- Implemented scalable Django microservices with dynamic database routing, Redis caching, and containerized workflows.
- Automated devops using Docker, Kubernetes, GitHub Actions, and Cypress for full CI/CD and infrastructure reproducibility.

### UCONN & State Street

May 2023 – Aug 2023

*Lead Project Developer*

- Co-led an equity sentiment forecasting initiative with State Street, modeling investor behavior from SEC filings and macroeconomic signals.
- Engineered a React and Google Cloud pipeline to ingest and normalize 10k+ daily SEC filings and news headlines.
- Developed BERT and VaderSentiment-based NLP workflows, reducing prediction MSE by 20% versus baseline.
- Documented insights via a detailed [project overview](#) and released the frontend [code](#), driving reproducibility and collaboration.

### Western Connecticut State University

Jun 2020 – Aug 2022

*Undergraduate Researcher*

- Designed and implemented a decentralized blockchain prototype with post-quantum security features using wavelet-based steganography and optical encryption.
- Built cryptographic data encoding and multiresolution signal processing tools in MATLAB; integrated custom transforms to enhance system resilience and performance.
- Published four papers and presented findings in [Gnoscience](#), [Springer](#), [ICSPIC](#), and [AMS](#)

## PROJECTS

### Live Systematic Equity Market-Making Signal — Smith Investment Fund

[SIF Website](#)

- Built a composite microstructure signal detecting informed order flow and adverse selection from bid-ask spreads, price impact, and abnormal volume — the same defenses a market maker uses to protect quotes.
- Flagged rising information asymmetry in **Silicon Valley Bank** weeks before its collapse; cleanly separated every stressed bank from every healthy one across a six-name universe.
- Deployed live via **Alpaca** on a \$50K portfolio with automated position sizing and stop-loss controls, achieving a **1.8 Sharpe** over 18 months.

## TECHNICAL PROFICIENCIES

**Languages:** Python (NumPy, Pandas, StatsModels), C/C++, Rust, SQL, MATLAB, OCaml, JavaScript/TypeScript, Java

**Quant & ML Libraries:** PyTorch, TensorFlow, Scikit-learn, XGBoost, PyPortfolioOpt, TA-Lib, yfinance

**Frameworks & Infrastructure:** Django, FastAPI, Flask, React, Docker, Kubernetes, Git, Google Cloud Platform, PostgreSQL, Redis